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ChE 210 Quiz #4

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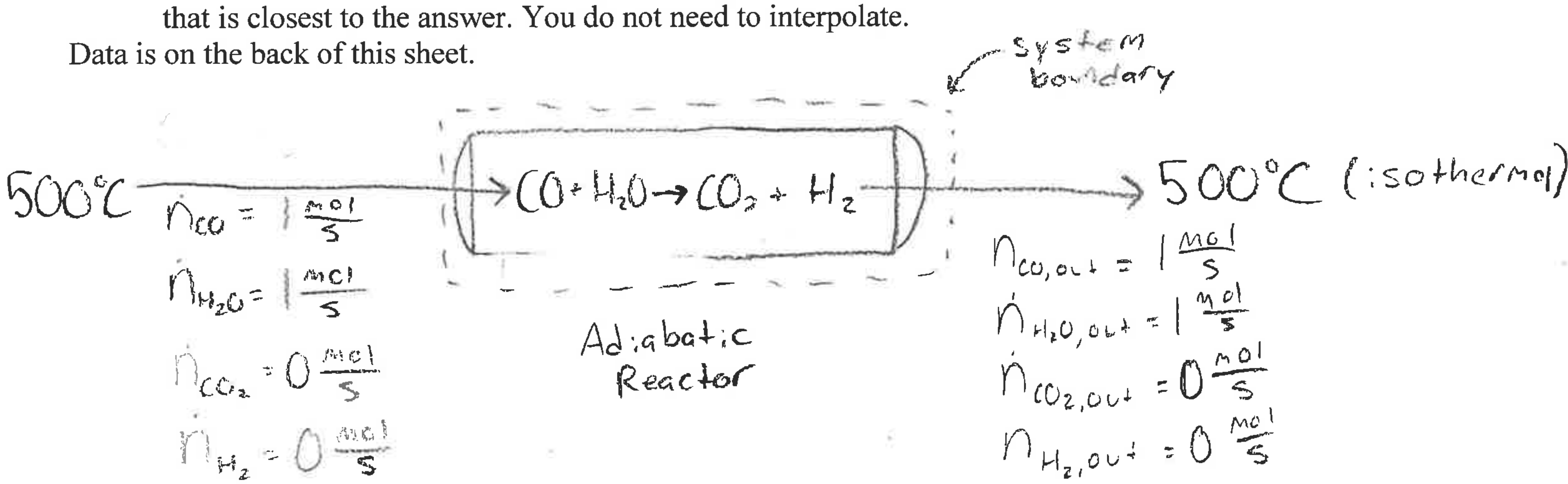
The gas phase reaction $CO + H_2O \leftrightarrow CO_2 + H_2$ takes place in an equilibrium reactor. The feed to the reactor enters at 500°C with a 1 mol/s of CO and 1 mol/s of H_2O and no CO_2 or H_2. The equilibrium relation for the reaction is:

$$64 = \frac{y_{CO2}\,y_{H2}}{y_{CO}\,y_{H2O}}$$

(a) Calculate the species molar flow rates of the exit stream in mol/s.
(b) Calculate the heat exchanged with the surroundings in kJ/s if the reactor is isothermal.
(c) Estimate the product gas temperature in °C if the reactor is adiabatic. Give the temperature in the table that is closest to the answer. You do not need to interpolate.

Data is on the back of this sheet.



a) $\dot n_{CO,out} = \dot n_{CO,in} - \dot\xi \longrightarrow \dot n_{CO,out} = 1\frac{mol}{s} - \dot\xi$

$\dot n_{H_2O,out} = \dot n_{H_2O,in} - \dot\xi \longrightarrow \dot n_{H_2O,out} = 1\frac{mol}{s} - \dot\xi$

$\dot n_{CO_2,out} = \cancel{\dot n_{CO_2,in}}^{0} + \dot\xi \longrightarrow \dot n_{CO_2,out} = \dot\xi$ $\quad \dot\xi = ?$

$\dot n_{H_2,out} = \cancel{\dot n_{H_2,in}}^{0} + \dot\xi \longrightarrow \dot n_{H_2,out} = \dot\xi$

b) $Q = \dot n \Delta \hat H_r^\circ$

$\hat H_r^\circ = \sum_{out} V_{i,out} \hat H^\circ_{f,out} - \sum_{in} V_{i,in} \hat H^\circ_{f,in}$

$\dot\xi$ $\quad$ CO_2 $\quad H_2 \quad CO \quad H_2O$

$(1)(21.34) + (1)(13.83) - (1)(14.38) + (1)(17.01) \longrightarrow \boxed{37.8\ \frac{kJ}{mol}}$

$(1)(\quad) + (1)(\quad) \longrightarrow (1)(\quad) + (1)($

missing factor of $\dot\xi$.

Table B.8 Specific Enthalpies of Selected Gases: SI Units

			$\hat{H}$(kJ/mol)				
		Reference state: Gas, P_{ref} = 1 atm, T_{ref} = 25°C					
T	Air	O_2	N_2	H_2	CO	CO_2	H_2O
0	−0.72	−0.73	−0.73	−0.72	−0.73	−0.92	−0.84
25	0.00	0.00	0.00	0.00	0.00	0.00	0.00
100	2.19	2.24	2.19	2.16	2.19	2.90	2.54
200	5.15	5.31	5.13	5.06	5.16	7.08	6.01
300	8.17	8.47	8.12	7.96	8.17	11.58	9.57
400	11.24	11.72	11.15	10.89	11.25	16.35	13.23
500	14.37	15.03	14.24	13.83	14.38	21.34	17.01
600	17.55	18.41	17.39	16.81	17.57	26.53	20.91
700	20.80	21.86	20.59	19.81	20.82	31.88	24.92
800	24.10	25.35	23.86	22.85	24.13	37.36	29.05
900	27.46	28.89	27.19	25.93	27.49	42.94	33.32
1000	30.86	32.47	30.56	29.04	30.91	48.60	37.69
1100	34.31	36.07	33.99	32.19	34.37	54.33	42.18
1200	37.81	39.70	37.46	35.39	37.87	60.14	46.78
1300	41.34	43.38	40.97	38.62	41.40	65.98	51.47
1400	44.89	47.07	44.51	41.90	44.95	71.89	56.25
1500	48.45	50.77	48.06	45.22	48.51	77.84	61.09

c) Adiabatic = $\dot{Q} = 0$ No Heat Transfer